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Game-supported cognitive strategy training for slowed information processing speed after acquired brain injury: study protocol for a randomised controlled trial

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ABSTRACT

Introduction Many individuals with acquired brain injury tend to experience problems with slowed information processing speed (IPS). A potentially beneficial and cost-effective supplement for cognitive rehabilitation of impaired IPS may be the implementation of serious gaming that focuses on compensatory learning as part of cognitive training. However, most digital platforms used during cognitive rehabilitation focus on restoring cognitive function and evidence for skill transfer from digital practice to everyday life is lacking. This study aims to investigate the efficacy of a game-supported cognitive strategy training. The training combines a well-validated time pressure management cognitive strategy training, targeting slowed IPS, with a novel game and a mobile application. The game-supported training focuses on the generalisation of strategy-use to untrained tasks in everyday life.

Methods and analysis The study is designed as a randomised controlled trial in which the experimental group (Karman Line — Tempo module: an 8-week game-supported cognitive strategy training) will be compared with an active control group (CogniPlus training: an 8-week computerised cognitive function training). Data from 60 individuals with acquired brain injury (30 per group, ages between 16 and 75) will be collected at baseline (T0), post-treatment (T1) and at 3-month follow-up (T2). The primary outcome measure is an objective assessment of compensatory strategy use in an untrained experimental task. The secondary outcome is the attainment of trained and untrained treatment goals assessed by goal attainment scaling. Pre-training and post-training data will be analysed using a 2×2 repeated measure analysis of variance.

Ethics and dissemination This study has been approved by the medical review ethics committee CMO Region Arnhem and Nijmegen (NL74818.091.20) and is registered in the Netherlands Trial Register. Research findings will be published in peer-reviewed journals and presented at conferences.

Trial registration number NL9437; The Netherlands Trial Register.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This study is the first to investigate a compensatory approach in serious gaming, combined with a scientifically validated cognitive training, as an intervention for slowed information processing speed (IPS) in acquired brain injury (ABI).
⇒ The design of this study is an adequately powered randomised controlled trial in which block randomisation and an active control group are used to control for confounding effects of computer training in a rehabilitation setting.
⇒ The outcome variables of this study measure the generalisation of the intervention to everyday life, especially to the level of activities and participation, in addition to improvement on neuropsychological tests.
⇒ Few studies exist that investigate digital strategy-oriented treatment for individuals with ABI and slowed IPS, therefore, power calculation was limited and based on one previous study that found high effect sizes.
⇒ The 3-month follow-up of this study is limited to online, self-administered questionnaires to place minimal burden on the participants.

INTRODUCTION

Background

Individuals with acquired brain injury (ABI) referred for outpatient rehabilitation often experience difficulties with processing and retaining information due to slowed information processing speed (IPS).[1] They may experience externally observed slowness (as noticed by others or measured by neuropsychological tests) and problems with information processing in everyday activities.[2] Consequently, they may experience problems such as fatigue, lowered mood and irritability.[3] The high frequency and ubiquity of these complaints and their impact on the lives of individuals with ABI make slowed IPS a prime target for cognitive rehabilitation.

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Cognitive rehabilitation in ABI has traditionally focused on the restoration of 'lost' cognitive functions, which only has limited effectiveness.[4] Compensatory strategy training, however, is more promising as it aims to improve the task performance of individuals within their possibilities. The goal of this type of training is to teach strategies to overcome cognitive deficits and support the performance of everyday activities.[5–7] Compensatory strategy training, as opposed to restorative function training, is currently widely used in the cognitive rehabilitation of people with ABI.[8]

The use of serious gaming in cognitive rehabilitation

While compensatory strategy training may be an effective approach for improving performance in daily life tasks, practising this type of training in face-to-face conditions under the guidance of a cognitive trainer is time-consuming and, thus, costly.[8 9] Incorporating elements from serious games (SG) or computerised cognitive training (CCT) in compensatory strategy training may reduce the need for constant supervision as it allows for guided home practice. We consider the label SG to denote either custom-designed experimental games or commercially designed games with an educative or health-improving approach.[10] The label CCT refers to cognitive training programmes that integrate game-like features to boost skills or cognitive functioning by repetitively performing demanding tasks over time.[11] Most commercially available SG and CCT, such as Lumosity, HAPPYNeuron, Rehacom or CogniPlus, focus on restoring cognitive functions rather than compensation. SG gained popularity due to the attractive and realistic simulation of everyday situations, with tasks that adapt their difficulty to the individual's level.

Several studies on the effectiveness of cognitive interventions for ABI have included SG and CCT.[10 12 13] A recent review by Ong, Weibin and Vallabhajosyula[10] has shown that adding SG to regular cognitive training improves rehabilitation outcomes when it aims to ameliorate the impact of neurological conditions. The implementation of SG in treatment allows for an objective assessment of the individual's progress[14] and enables more individualised treatment programmes tailored to specific rehabilitation needs.[15] Moreover, this implementation leads to better adherence to rehabilitation regimens due to the possibility of guided home practice[16] and improved motivation and engagement.[17]

In contrast, studies on CCT in individuals with ABI have shown mixed results concerning their efficacy and effectiveness in cognitive rehabilitation.[12 13] Most studies found little or no transfer of benefits to non-trained skills or daily functioning,[18 19] likely due to the focus of the CCT programmes on restoring cognitive function rather than compensation for cognitive problems. The beneficial effects of CCT are mostly limited to improvement in trained tasks. In addition, positive effects are short-lasting[20] and many studies have flawed methodology.[13 21 22]

A systematic review by Sigmundsdottir *et al*[12] provides several recommendations for improving investigations in the field of cognitive rehabilitation. The authors propose applying a more rigorous methodology, such as using more rigid RCT designs, active computer-based control groups and appropriate activity and participation outcome measures (rather than cognitive test improvement). Combining computer training with high-dose therapy-assisted interventions may be more effective than isolated CCT or SG, especially when the intervention combines training-specific cognitive functions with education on how to apply compensatory strategies to everyday life.[23–25]

Game-supported cognitive strategy training

The current project aims to investigate the Karman Line — Tempo module, a newly developed game-supported cognitive strategy training for brain-injured individuals who experience slowed IPS.[26] The Karman Line — Tempo module combines a compensatory cognitive strategy training with a strategy-oriented game (Tempo game) and a mobile application (Tempo tool). Figure 1 presents an overview of the different components of the Tempo module. Unlike existing SG and CCT, our game set aims to support a shortened version of an already validated face-to-face compensatory strategy training, namely, time pressure management (TPM) training.[6 27] In TPM training, individuals become familiar with applying the five-step TPM strategy for the execution of everyday activities. They first identify when they experience time pressure during the execution of a task and then compensate for slowed IPS through strategies to prevent or reduce this time pressure. The individual learns how to use the TPM strategy under the guidance of a trained therapist. Adding the game set to the TPM training enables guided home practice. The Tempo game allows for practising the TPM method in an interactive game and the Tempo tool assists during the execution of daily activities by creating a set-by-step plan according to the TPM method.

This protocol paper describes the study design used to investigate the efficacy of the Karman Line — Tempo module, a game-supported cognitive strategy training for people with ABI and slowed IPS. The main focus of this study is to investigate possible generalisation of compensatory strategy use in everyday situations. We hypothesise that implementing the Tempo module will lead to a more effective generalisation of compensatory strategies to untrained tasks in individuals with ABI relative to an active control group who will receive computerised cognitive function training. We expect that the Tempo module contributes to improved treatment of slowed IPS in individuals with ABI.

METHODS AND ANALYSIS

Study population

The study population consists of individuals with ABI referred for outpatient cognitive rehabilitation. A total of

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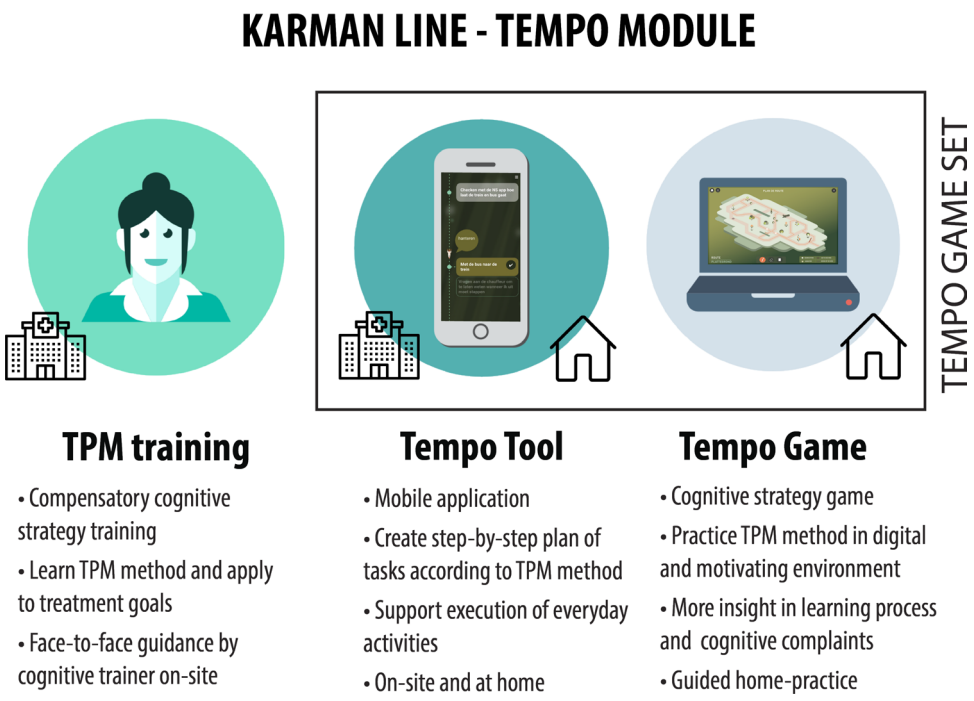


Figure 1 Schematic overview of the Karman Line — Tempo module. A cognitive training (TPM training, presented on the left side), focused on learning compensatory strategies for slowed information processing speed, is supported by the Tempo game-set (presented on the right side). The Tempo game-set includes a computer game (Tempo game) and a mobile application (Tempo tool). TPM, time pressure management.

60 participants will be enrolled in the study. To be eligible to participate, a person must meet all of the following inclusion criteria: (A) suffering problems due to slowed IPS (as assessed by a Mental Slowness Questionnaire (MSQ)[28] score ≥13) following ABI of non-progressive nature (ie, traumatic brain injury, stroke); (B) a minimal postonset time of 3 months after brain injury; (C) age between 16 and 75 years and (D) living independently at home. Exclusion criteria of the study are: (A) inability to speak/understand the Dutch language; (B) a history of severe psychiatric problems that affect current participation in the study (ie, severe depression, post-traumatic stress disorder); (C) neurodegenerative disorders; (D) substance abuse; (E) severe cognitive comorbidity that affects participation in the study (eg, dementia), (F) inability to look at a computer screen for more than 15 min and/or operating a keyboard and/or operating a computer mouse; (G) aphasia and (F) visual neglect.

Individuals will be recruited from five locations of three rehabilitation centres in the Netherlands: the outpatient clinic and the Department of Neurorehabilitation of Klimmendaal rehabilitation specialists (location Arnhem, location Ede and location Apeldoorn, the Netherlands); the Department of Neurorehabilitation of Military Rehabilitation Centre Aardenburg in Doorn, the Netherlands and the Department of Neurorehabilitation of Rehabilitation Centre Friesland (location Leeuwarden and location Beesterzwaag, the Netherlands). The study has been approved by the medical review ethics committee CMO Region Arnhem and Nijmegen (NL74818.091.20) and is registered at the Netherlands Trial Register.

Study design

The study will be a randomised controlled trial (RCT) designed to determine the efficacy of the Karman Line — Tempo module: a game-supported strategy training for individuals with ABI aimed at improving compensatory strategy use for slowed IPS in daily life activities.[26] The experimental group (8-week Karman Line — Tempo module) will be compared with an active control group (8-week CogniPlus training: a computerised cognitive function training). We followed the recommendations of the systematic review on CCT in neurorehabilitation by Sigmundsdottir *et al*[12] in the design of our RCT.

Participants will be randomly assigned using variable block randomisation stratified by gender and age, using Castor Electronic Data Capture (EDC).[29] A researcher will administer measurements before (T0: baseline) and after the 8-week intervention (T1: post-treatment).The treatment will be given by a trained cognitive trainer or neuropsychologist. A follow-up measurement (T2) consisting of questionnaires will be obtained 3 months after treatment. The intervention is single-blinded: investigators were not blinded for group allocation as the nature of the outcome measures ensured a non-biased assessment of performance (eg, computer-obtained measurements). Figure 2 presents a schematic overview of the study.

Recruitment procedures and consent

Potential participants are informed about the study and asked to fill in the MSQ before a first intake at the rehabilitation centre. Informed consent will be obtained if the individual meets the inclusion criteria and decides

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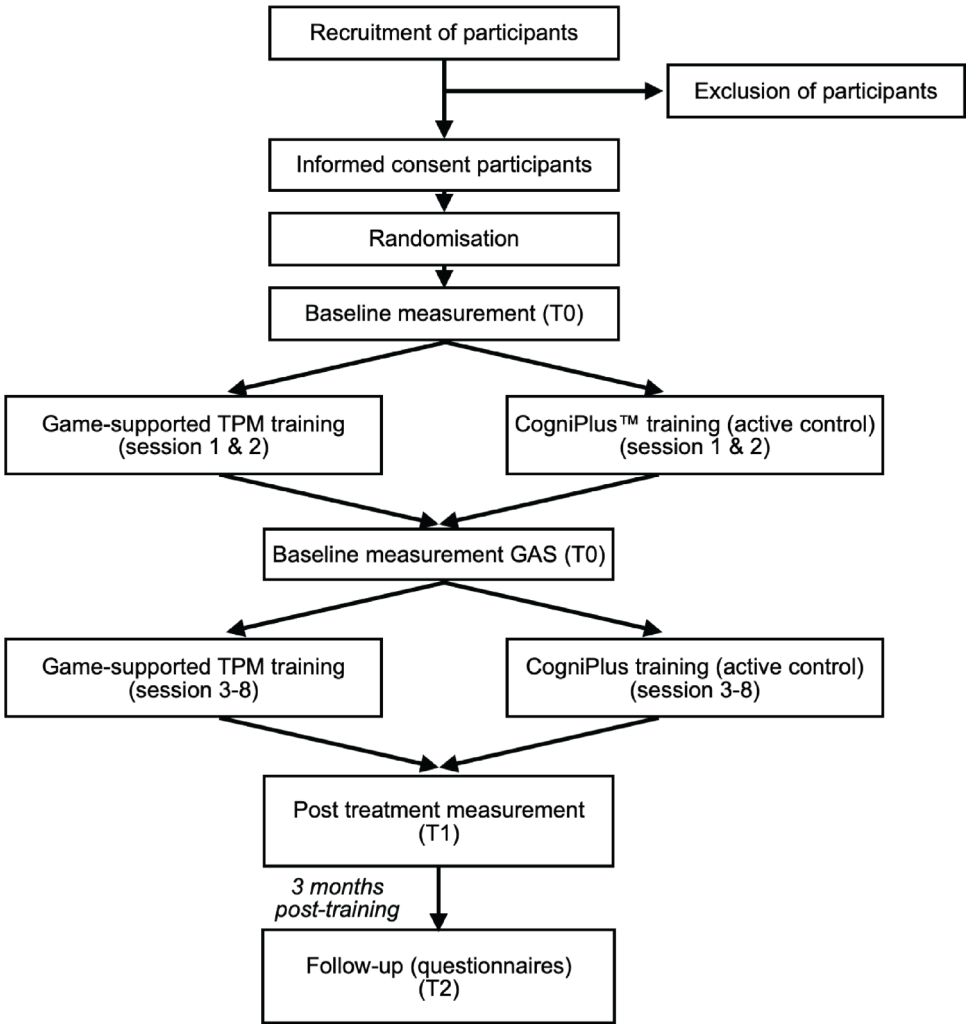


Figure 2 Schematic overview of the study design. GAS, goal attainment scaling; TPM, time pressure management.

to participate after a 1-week decision period. The study information and consent form provided to the participant are presented in online supplemental appendixes 1 and 2. Recruitment will take place between December 2021 and July 2023.

Interventions

Karman Line — Tempo module: an experimental game-supported strategy training for slowed IPS

The Karman Line — Tempo module[26] is based on the TPM training, a scientifically validated and successfully implemented treatment for individuals who experience slowed IPS.[6 27 30] The TPM training focuses on the acquisition of compensatory strategies that allow individuals to recognise and deal with moments of time pressure in everyday situations. The five-step TPM strategy is based on the general self-instructional idea of 'Let me give myself enough time to do a task'.[6 30] Participants are guided by five main questions in the performance of a task while compensating for slowed IPS (table 1)

The Karman Line — Tempo module combines a shortened version of the conventional TPM training with a custom-designed game set. The training involves eight sessions at the rehabilitation centre (see figure 1 and table 2). Each therapist-guided training session is 1–1.5 hours with a recommended frequency of one session per week. The total duration of the training sessions is 8.5 hours. The game set consists of an innovative cognitive strategy game (Tempo game) and a mobile application (Tempo tool). The Tempo game is played on a computer and allows for guided home practice of the TPM method in a digital environment. The content of the game follows the pace of the face-to-face TPM training: the individual will only be able to apply the strategies they have learnt during the face-to-face training to ensure an optimal learning curve. The Tempo game has 24 levels with an approximate playing time of 15 min per level. The approximate total playing time of the full game is 6–18 hours. The Tempo tool is used on a smartphone to create a step-by-step plan according to the five-step TPM strategy and directly apply compensatory strategies in daily life. The Tempo game set is, thus, not intended as a replacement for the therapist's role or the treatment itself but as an addition to the compensatory cognitive strategy training. The duration of the game-supported TPM training (Tempo module) is 8.5 hours as opposed to the 10 hours of the original TPM training,[30] which is

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Table 1 Overview of the five-step TPM strategy

Steps	Description	Main objective
Step 1: Describe the task	1. Give a short task description. 2. Divide the task into smaller steps.	Recognise time pressure in the task at hand.
Step 2: Create an action plan	1. Indicate which steps do not induce time pressure. 2. Indicate which steps may induce time pressure. 3. Which steps can you take before starting the task? 4. Indicate what you can do during 'calm moments' (moments with no activity between steps).	Prevent as much time pressure as possible.
Step 3: Create strategies and an emergency plan	1. Write down strategies to manage time pressure for each step that may induce time pressure. 2. Create an emergency plan for unexpected events.	Deal with time pressure as quickly and effectively as possible.
Step 4: Execution of the action plan	1. Be aware of moments that may induce time pressure. 2. Apply strategies to manage time pressure during steps that may induce time pressure. 3. When an unexpected event occurs: apply the emergency plan.	Monitor yourself while using the TPM strategy.
Step 5: Task evaluation	1. Evaluate for each executed step whether you implemented the action plan successfully. 2. What can you improve for the steps that did not go as planned?	Get insight into the implementation of the TPM strategy during task execution.

TPM, time pressure management.

made possible by the engaging nature of the game set and digitally guided home practice.

The Tempo game set was co-created with patient volunteers with brain injury, therapists and neuropsychological experts to ensure an effective and user-friendly addition to the treatment. All parties provided feedback on the tool and the individual game levels throughout the development process. Feedback sessions were organised to explore the player's experience, ensure that the game and tool matched the needs and learning objectives of the target group and improve the usability and design of the game and tool.

The Karman Line — Tempo module consists of three main stages. The first stage of the training (sessions 1 and 2) focuses on improving insight into slowed IPS and recognising time pressure. The Tempo game will be used during the first phase to practice creating a step-by-step plan of everyday activities and recognise the steps that invoke time pressure. In the second stage, the therapist explains the TPM method, and three personal treatment goals are set up (sessions 3–5). The participants will play the Tempo game at home to practice the TPM method and apply compensatory strategies in several situations, such as notetaking during a conversation and navigating a maze. Three personal treatment goals are created using Goal Attainment Scaling (GAS).[31 32] Two goals will be trained during the training (using the Tempo tool) and the third goal will remain untrained to investigate whether training effects generalise to untrained tasks. The third stage focuses on generalising the TPM method to functioning in everyday life. Again, the participants will play the Tempo game at home to practice the TPM strategy under more difficult circumstances and apply the Tempo tool to new activities.

Active control group: CogniPlus training

The active control group will receive the SPEED module of the CogniPlus training programme,[33] a CCT programme with a scientific background that aims to train IPS.[34] No compensatory strategies are taught during this intervention. The CogniPlus training programme is based on the restorative approach and, thus, on the assumption that underlying cognitive impairment can be restored, or at least move in the direction of normality, through cognitive exercises.[35]

The active control group receives eight 1–1.5-hour sessions at the rehabilitation centre (see table 3). The total duration of the treatment is 8.5 hours, and the recommended frequency of the training is one session per week. The first two sessions focus on providing psychoeducation and setting three personal goals according to the GAS scales. From sessions 3–7 (phase 2), the SPEED module is self-administered on-site while the therapist is in the room to assist if necessary. During session 8 (phase 3), the individual's final performance on the training is discussed.

Primary outcome measure

The primary outcome measure is the Virtual Meeting Task (VMT), an objective assessment of strategy use in an untrained computerised task (see figure 3 and table 4). The VMT examines strategy planning, strategy execution and task execution in a pre-recorded video call. The task was designed for adults who are living in the Netherlands; therefore, the conversations in

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Table 2 Overview of the Karman Line — Tempo module: a game-supported cognitive strategy training

Karman Line — Tempo module		
Phase 1 ► Improve insight in slowed IPS, ► Recognise situations with time pressure	**Session 1** Information and insight Duration: 1 hour	► Introducing participants to the game-supported TPM training. ► A brief overview of the programme's purpose and each session's content. ► Provide psychoeducation on slowed IPS and time pressure, secondary problems due to slowed IPS (emotional changes, fatigue, forgetfulness), and the relationship between slowed IPS and impaired performance in everyday life. ► Explain the objective of the Tempo game: practice the time pressure management strategies at home in several game levels. ► Explain the objective of the Tempo tool: create a step-by-step plan for the execution of activities and use the tool to practice these activities at home. ► No game or tool
	Session 2 Setting up goals Duration 1.5 hours	► Define three personal treatment goals (scored on a Goal Attainment Scale) according to the SMART principle, with support from the practitioner. ► Two of these goals will be practised during the treatment, and the third goal remains untrained to investigate whether training effects transfer to untrained tasks. ► Tempo game (level 2.1–2.4): Introductory. The player learns how to recognise time pressure in four different scenarios. For example, the avatar is lost in a park and has to ask for directions. The player may experience time pressure when listening to detailed navigational instructions. Total duration games: 0.5–1 hour.
Phase 2 ► Acquisition of the TPM method	**Session 3** Introduction TPM method Duration: 1 hour	► Introduce participants to the TPM method aimed at recognising, preventing and dealing with time pressure. ► Explain that the participant's IPS does not return to premorbid levels but that they can use the TPM strategy, a five-step strategy, to compensate for their slowed IPS. ► Tempo game (level 3.1–3.4): Easy difficulty. The player practises with the TPM method by setting up an action plan according to the five-step TPM strategy and applying different compensatory strategies to prevent and deal with time pressure in different situations. Games include making notes during conversations or navigating through a maze. Total duration games: 1–2 hours
	Session 4 First treatment goal Duration: 1 hour	► Create a step-by-step plan for the first personal treatment goal following the five-step TPM strategy under the practitioner's guidance. ► Tempo tool: The participant uses the Tempo tool on their smartphone to practise the created step-by-step plan at home. ► Tempo game (level 4.1–4.4): Medium difficulty. The assignments in the game become increasingly difficult by, for example, increasing the number of task steps, introducing the application of an emergency plan and increasing the speed of the games. For example, the player has to plan a route and navigate through a maze while distractions and difficult decisions are presented at a more rapid tempo. Total duration games: 1–2 hours
	Session 5 Second treatment goal Duration: 1 hour	► Create a step-by-step plan for the second personal treatment goal following the five-step TPM strategy under the practitioner's guidance. ► Tempo tool: The participant uses the Tempo tool on their smartphone to create the step-by-step plan together with the practitioner and practice it at home. ► Tempo game (5.1–5.4): Medium difficulty. Level of difficulty increases by increasing the number of task steps and introducing distractions during a game. For example, distractions (eg, interruption by another avatar) are introduced when the player has to listen to a conversation or prepare a meal. Total duration games: 1–2 hours

Continued

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Table 2 Continued

Karman Line — Tempo module		
Phase 3 ► Generalisation	**Session 6** Generalisation to everyday life situations Duration: 1 hour	► Provide psychoeducation on the generalisation of skills to everyday life. The practitioner explains that therapists cannot possibly treat all the problems and tasks participants will encounter in their surroundings. Therefore, generalising the TPM strategy from the rehabilitation to the home setting is essential. ► Create a step-by-step plan for a new situation according to the 5-step TPM strategy for a new situation in the Tempo tool. ► Discuss and practice situations with time pressure from the participant's daily life. ► Tempo tool: The participant applies the tool to new situations in everyday life and is encouraged to practice these tasks at home. ► Tempo game (level 6.1–6.4): Advanced difficulty. In these final levels, the player receives less feedback and guidance from the game, the speed of the games increases, more distractions are introduced, and independence of the player's decisions is encouraged. Total duration games: 1–2 hours
	Session 7 Tempo tool in everyday life Duration: 1 hour	► The participant practises various situations with time pressure under more distracting and challenging conditions. ► Tempo tool: The participant can independently create new step-by-step plans in the tool and is encouraged to practice situations with time pressure at home.
	Session 8 Evaluation & final session Duration: 1 hour	► Evaluate achievement of personal treatment goals (according to the GAS) ► Evaluate the experience of the overall treatment Encourage the participant to continue using the five-step TPM strategy and the Tempo tool daily.

GAS, goal attainment scaling; IPS, information processing speed; SMART, Specific, Attainable, Reasonable, Timely; TPM, time pressure management.

the task are in Dutch. The participants will perform parallel, randomised versions of the VMT at baseline level (T0) and at an 8-week follow-up (T1). In the VMT, the participants are instructed to partake in a pre-recorded virtual meeting with four actors and answer questions about the content of the conversation. Participants can use strategies to prevent time pressure (e.g., taking notes) and to manage induced time pressure by selecting strategy buttons that elicit a response (e.g., asking for repetition). After each scene, the participants answer questions on the content of the conversation. Performance in the VMT will be scored on the following aspects: (1) strategy planning (number of correctly chosen strategies that can prevent time pressure), (2) strategy execution (number of correctly chosen strategies that help manage time pressure) and (3) task performance (number of correct reproductions of information points). After completing the task, the participants perform a self-rating questionnaire, which measures personal insight into the subjective experience of perceived task performance and cognitive state. Construct validity, test–retest reliability, parallel-form reliability and ecological validity of the experimental task will be investigated in a study that runs simultaneously with the study described in this protocol.

The VMT was explicitly designed for this study to investigate the application and generalisation of compensatory strategies in a simulated everyday setting. While previous studies only focus on measuring cognitive function, our study follows the recommendations of Sigmundsdottir *et al*[12] by using this task to measure possible improvement at the level of activities and generalisation of skills to everyday life.

Secondary outcome measure

The secondary outcome measure is a standardised GAS score of three personal treatment goals.[31 32] GAS is an individualised method used in brain injury rehabilitation to evaluate to what extent individual treatment goals are achieved. The achievement of each goal will be measured on a 5-point scale.[36 37] Each individual has specific treatment goals, which are scored standardised to allow statistical analysis. For the present study, three personal treatment goals will be identified that relate to situations in which the individual experiences problems due to slowed IPS. Acceptable treatment goals can be subdivided into multiple steps and are defined according to the SMART principle (Specific, Attainable, Reasonable, Timely).[32 38]

Patient-reported outcome measures, such as GAS, are considered increasingly important in neurorehabilitation

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Table 3 Overview of the CogniPlus training: a computerised cognitive training

CogniPlus training		
Phase 1 ► Improve insight in slowed IPS ► Introduction CogniPlus training programme	**Session 1** Information and insight Duration: 1 hour	► Introducing participants to the CogniPlus training programme. ► A brief overview of the programme's purpose and each session's content. ► Provide psychoeducation on slowed IPS and time pressure, secondary problems due to slowed IPS (emotional changes, fatigue, forgetfulness), and the relationship between slowed IPS and impaired performance in daily life. ► Explain the objective of the SPEED module of the CogniPlus training programme, that is, to improve problems due to a slowed IPS by practising computer tasks to improve the speed of information processing.
	Session 2 Setting up goals Duration: 1.5 hours	► Create three personal treatment goals (scored on the GAS) according to the SMART principle, with support from the practitioner. ► Two of these goals will be practised during the treatment, and the third goal remains untrained to investigate whether training effects transfer to untrained tasks.
Phase 2 ► CogniPlus training programme	**Session 3–7** Training with the SPEED module Duration: 1 hour per session	► The participant will spend 1 hour per week practising different tasks of the SPEED module using three scenarios: ordering objects in ascending order, sorting objects on shelves, and avoiding obstacles on a track. These scenarios impose time pressure, and the participant is encouraged to decrease their reaction time. The participant can switch between scenarios in a self-directed manner. ► The practitioner will be present in the room to help the participant if they have any questions and to track their progress. ► After the participant finishes practising the tasks, the practitioner will reflect with the participant on the performance based on the training session results.
Phase 3 ► Final results SPEED module	**Session 8** Evaluation and final session Duration: 1 hour	► The participant plays all scenarios from the SPEED module as a post-treatment measure. ► Evaluate the participant's progress throughout the training. ► Evaluate the achievement of personal goals (according to the GAS) ► Evaluate the experience of the overall treatment.

GAS, goal attainment scaling; IPS, information processing speed.

research and are critical for developing patient-centred approaches for new treatments.[39 40] We included GAS to ensure that the personal value of the treatment is assessed for individuals with ABI and to ultimately improve treatment satisfaction and the quality of care.

Additional study parameters

Demographic measures include age, sex, education level, type of brain injury, time after injury and slowed IPS severity. IPS is assessed by measuring reaction time (ms) using the Alertness subtest of the Test for Attentional performance (TAP, V.2.3.1).[41] A clinically significant slowness in IPS is considered when individuals score 1.5 SD below the norm on this test.

Other study parameters include self-rating questionnaires measuring compensatory strategy use in everyday life (Time Pressure Questionnaire[27]), participation in daily life (Utrecht Scale for Evaluation of Rehabilitation-Participation[42]; USER-P), subjective motivation (Motivation for Traumatic Brain Injury Rehabilitation Questionnaire[43 44]), subjective fatigue (Dutch Multifactor Fatigue Scale[45]; DMFS) and insight and reported severity of mental slowness (Mental Slowness Questionnaier; MSQ[28]). A selection of questionnaires (Time Pressure Questionnaire, MSQ, USER-P, DMFS) will be administered at 3-month follow-up to assess the long-term effects of the training.

A neuropsychological test battery will be administered to assess IPS and other aspects of cognitive functioning at baseline. For this purpose, subtests of the TAP (V.2.3.1)[41] will be administered. Specifically, the Alertness subtest to assess general processing speed, the Sustained Attention subtest to assess the longer term maintenance of attention, the Working Memory subtest to assess the ability to update the working memory and the Go/No-go subtest to assess cognitive control. The Attention Network Test[46] (ANT) is used to assess the severity of slowed IPS. The Brixton Spatial Anticipation test[47] will be administered to assess set-shifting and perseveration. The Dutch version of

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Figure 3 Overview and task outcomes of the Virtual Meeting Task for objective assessment of compensatory strategy use. Note: The people presented in this figure are professional actors who provided explicit consent for the inclusion of this image in scientific journals. The names in the figure are Dutch pseudonyms depicting a specific role in the task and do not reflect the real names of the actors.

the National Adult Reading Test[48 49] (NART) is included to estimate premorbid intelligence quotient (IQ). A selection of neuropsychological tests will be repeated post-treatment to control for potential non-specific recovery. An overview of the measurements is presented in table 5.

Data will be obtained to evaluate the game performance of the individuals who receive the Karman Line — Tempo module. For example, the player's reaction times and the number of correct responses will be collected during each individual game level. Based on this data, variables are created to estimate strategy use, understanding of the game, attention, fatigue, perseveration and motivation. Predictions are made by a general probabilistic scaling method underlying the game that can estimate variables based on their posterior probabilities.[50] The validity of this diagnostic network will be investigated in another study that runs simultaneously with the study described in this protocol.

Data management

Study data will be collected and managed using Castor EDC[29] hosted at the Donders Institute for Brain, Cognition and Behaviour, Nijmegen, the Netherlands. Castor EDC is a secure, web-based software platform that supports data capture for research studies. Participants will be identified using a study-specific identification code. One researcher will keep a separate participant identification code list in the protected project folder at the sponsor's site (rehabilitation centre Klimmendaal, Arnhem, the Netherlands) that matches the study-specific identifying codes with the participants' names. Paper documents are kept in a locked safe at the sponsor's site.

Patient and public involvement

We involved individuals with ABI and members of the public in several stages of the development of the game-supported treatment. The Tempo game set was co-created

Table 4 An overview of the task outcomes of the Virtual Meeting Task for objective assessment of compensatory strategy use

Action button	Description
Pause/resume	By pressing this button, all speakers stop talking. To continue, press the same button again.
Repeat	Press this button to repeat the speaker's last line(s).
Slower	The current speaker repeats the last line(s) slower by pressing this button. This button only works if the speaker is talking too fast.
Focus speaker	By pressing this button, the speakers stop talking at the same time. This button only works when multiple speakers are talking at the same time.
Focus topic	Press this button to redirect the conversation back to the relevant topic. This button only works if the speakers stray from the main topic.
Question	Press this button to ask a specific question. Only use this button if the assignment at the beginning of the scene is to ask a question when a particular topic is brought up.

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Table 5 Inclusion and outcome measures

Type	Measurement	Inclusion	Baseline	Post-treatment	Follow-up
Primary outcome measure	Virtual Meeting Task (VMT)		x	x	
Secondary outcome measure	Goal Attainment Scaling (GAS)		x	x	
Neuropsychological tests	Test Battery for Attentional performance (TAP)				
	Alertness	x	x		
	Working memory	x			
	Sustained attention	x			
	Go/No-Go	x			
	Brixton Spatial Anticipation test		x		
	Attention Network Test (ANT)		x	x	
	National Adult Reading Test (NART)		x		
Questionnaires	Mental Slowness Questionnaire (MSQ)	x	x	x	x
	Utrecht Scale for Evaluation of Rehabilitation-Participation (USER-P)		x	x	x
	Strategy-use in individuals with slowed information processing speed (IPS) (Questionnaire 'Omgaan met tijdsdruk')		x	x	x
	Dutch Multifactor Fatigue Scale (DMFS)		x	x	x
	Motivation for Traumatic Brain Injury Rehabilitation Questionnaire (MOTQ)		x	x	

with volunteers with ABI, therapists and neuropsychological experts to ensure an effective and user-friendly addition to the treatment. All parties were asked to provide feedback on the tool and the individual game levels throughout the development process. Feedback sessions were organised to explore the experience of individuals with ABI, to ensure that the game and tool matched the needs and learning objectives of the target group and improve the usability and design of the game and tool. For example, the initial plan for the Tempo game set was to create a mobile game to practice the skills that were taught during cognitive treatment. After the feedback sessions, it became apparent that the volunteers preferred to view the games on a computer screen instead of a mobile phone. Moreover, volunteers preferred to have a separate mobile application to support their day-to-day tasks. This feedback led to the development of the Tempo game and Tempo tool. Other examples of feedback that led to incorporation in the game were the preference of 2D over 3D gameplay, the use of soothing colours and limited distraction in its visual design, the inclusion of text-to-speech and all feedback rounds on user experience that guided the usability and difficulty of the levels in the game.

Together with rehabilitation experts, we carefully assessed the implementation of the trial interventions to minimise the burden on individuals with ABI. We intend to inform the participants of the study's main results and involve individuals with brain injury in discussing appropriate methods to disseminate the research outcomes.

Sample size

A power calculation for the primary outcome measure was based on the results of a prior study that used a similar compensatory strategy task (the Information Intake task), study design and research population.[28] We based our expected standardised effect size of at least 0.66 on the effect size (Cohen's *d*) of the variable 'number of used strategies' of the Information Intake task. Using the software program G*power (V.3.1),[51] we computed that a selected sample of 60 participants (30 per group) will be required to reach a power of 0.8 (α=0.05).[52] A detailed description of the sample size calculation is found in online supplemental appendix 3.

Analysis of primary outcome measure

Pre-training and post-training data obtained in the VMT will be analysed using a 2×2 repeated measure analysis of variance (ANOVA) to evaluate the efficacy of the game-supported treatment compared with the active control group. The treatment group (two groups: game-supported TPM training and active control group) is used as a between-subject factor and the moment of measurement (two levels: pretreatment, post-treatment) as a within-subject factor. Three aspects will be measured by the experimental task (strategy planning, strategy execution and task execution). The individual aspects of the task will be transformed into standardised scores and grouped into a single compound measure score as a primary outcome measure.

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Analysis of secondary outcome measure

As a secondary outcome measure, pretraining and post-training GAS scores of the three (un)trained goals will be analysed using a 2×2 repeated measure ANOVA. The treatment group (two groups: game-supported training and active control group) is used as a between-subject factor and the moment of measurement (two levels: pretreatment and post-treatment) as a within-subject factor.

Analysis of additional study parameters

Correlations will be computed between moderator variables and the treatment effects (difference score post-treatment minus pretreatment). Regression analysis will be performed to investigate the relation between the measured outcomes of the game (error rates, reaction times) and baseline measures of neuropsychological assessment and questionnaires.

Path analyses will be performed to assess the effectiveness of the Bayesian network that underlies the Tempo game in creating predictions of cognitive values obtained in the game (e.g., decrease in attention). Predictive models will be evaluated to examine the relationships between the variables obtained in the Tempo game and the outcomes on neuropsychological tests. Four predictive models will be tested. For the first model, the game variable obtained for strategy-use was used to predict the strategy execution, task execution and strategy planning score of the VMT. The second model uses the game variable for attention as a predictor of the Alertness subtest and the Sustained Attention subtest of the TAP. The third model uses the game variable for fatigue as a predictor of the course of reaction time of the Alertness subtest of the TAP and the outcome of the DMFS questionnaire. For the fourth model, we used the game variable for perseverance as a predictor for perseverative errors on the Brixton Spatial Anticipation test.

Ethics and dissemination

This study has received ethical approval from CMO Region Arnhem-Nijmegen medical research ethics committee (NL74818.091.20) and is registered in the Netherlands Trial Register (Accessible via the Dutch Trial Register: https://clinicaltrialregister.nl/en/trial/22586). Written consent from the patient is obtained before participating in the study (see online supplemental appendix 2 for a translated example of the informed consent form). The study will be conducted according to the principles of the Declaration of Helsinki (64th WMA General Assembly, Fortaleza, Brazil, October 2013) and following the Medical Research Involving Human Subjects Act (Wet Medisch-wetenschappelijk Onderzoek met mensen; WMO). All participants in the active control group are offered the conventional TPM training at Klimmendaal Rehabilitation Centre after the completion of the study.

Handling of personal data will comply with EU privacy law, General Data Protection Regulation (GDPR) and the Dutch Act on Implementation of the GDPR. Donders Institute and Radboud University have established a data management infrastructure in which all relevant data will be stored in a central storage space. Anonymised research data will be copied to the central storage system.

A site firewall protects the processing of all data on computers. Access to study data on the central storage system is restricted to specifically defined user groups or roles. Access to data is managed by the researchers responsible for the data, and they decide to give access rights to other individuals when necessary or requested. Anonymised research data are archived on the Donders Institute Research data repository (RDM) located on the Radboud university campus (https://data.donders.ru.nl).

Personal data are stored separately from research data in password-protected databases and stored on protected project folders on the servers of Klimmendaal Rehabilitation Centre. The researcher will ensure that the participants' identities will be protected. In all documents, participants will be identified by an identification code only, that is, not by their names or any other feature through which participants can be identified. The researcher will keep a separate Subject Identification Code List (Pseudonymisation-Key-file), which matches identifying codes with the participant's name. The researcher will maintain critical files strictly confidential and preserved for a certain period of time, meeting with a demonstrable legitimate interest (according to GDPR). Subject anonymity will be maintained throughout all archiving steps.

The data will be stored and archived through the data management infrastructure of the institutes, which will be kept for at least 15 years after the last publication at the institutes (following guidelines for minimal-risk research). The researcher will archive essential documents in a study Master file.

Following the institutional policy that our RDM should be FAIR (Findable, Accessible, Interoperable and Reusable), research data are made public immediately following publication of the outcomes of this study. Pseudonymised experimental data will be shared as soon as data collection from this study has been completed and has appeared in a full journal publication. Moreover, the study protocol, informed consent form, statistical analysis plan and analytic code will be made available on publication. Data sharing will be done using the Donders Institute RDM (http://data.donders.ru.nl) following institutional and EU guidelines. The data can be accessed via the Data Sharing Collection of this repository for at least 15 years after the last publication at the institutes. Researchers who provide a methodologically sound proposal can request access for secondary analysis with the managers of the data sharing collection (Managers Data Sharing Collection: Professor R. Kessels, roy.kessels@donders.ru.nl; Dr D. Bertens, dirk.bertens@donders.ru.nl; AC Abelmann, amy.abelmann@donders.ru.nl). Secondary analysis is considered when the analysis has the purpose of achieving the aims in the approved proposal. To gain access, data requestors will need to sign a data use agreement. A data

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use agreement specifies the conditions for usage of the data, ensuring that local institutional guidelines as well as (inter)national (EU) law (GDPR) are met.

Dissemination among the academic community and general public will involve publications in international, peer-reviewed journals, presentations at leading national and international conferences, media appearances in (local) newspapers, on social media, online blogs and public events. The publication of this study will not be limited to positive findings.

DISCUSSION

This RCT aims to investigate the efficacy of the Karman Line — Tempo module, a game-supported cognitive strategy training for individuals with ABI who experience slowed IPS. This study aims to (1) examine the generalisation of strategy use as assessed in an untrained experimental task and (2) investigate individual subjective experience of achievement of trained and untrained treatment goals as measured by standardised GAS.

The main strength of this study is the novelty of the compensatory approach in the Tempo module. This study is the first to investigate a game-supported cognitive strategy training that implements novel custom-designed digital platforms with a compensatory approach for ABI rehabilitation. Existing CCT and SG interventions mainly focus on improving cognitive function, and previous research has shown limited generalisation of skills to everyday life activities.[10 18–20 35] The Tempo module overcomes these limitations by combining therapist-guided compensatory cognitive training sessions with a custom-made digital game (Tempo game) and a mobile application (Tempo tool) that enables home-guided practice. All components of the Tempo module focus on teaching individual's compensation strategies for slowed IPS and guiding them in applying these strategies in everyday life, providing them with a robust basis for long-term improvement.

Another strength is the study's experimental design, which incorporated the recommendations formulated in the review by Sigmundsdottir *et al*.[12] The authors criticised the insufficient methodological quality of previous research that used CCT as an intervention in ABI. Previous studies showed a weak methodical design and a lack of active control groups. We apply an adequately powered RCT design and implement the CogniPlus training programme as active computer-based control group. We also include outcome variables that measure the generalisation of the intervention, especially the level of activities and participation, in addition to improvement on neuropsychological tests.

A possible limitation is that the 3-month follow-up measurement of this study includes self-rated questionnaires for feasibility reasons. No primary, secondary and neuropsychological measurements were included. In addition, since few studies exist that investigate digital strategy-oriented treatment for individuals with ABI and slowed IPS, the sample size calculation of the study is based on the outcomes of the study of Winkens *et al*.[28] The researchers used a task that is similar to our primary outcome measure and found a relatively large effect size (Cohen's *d* of 0.66), which we used in our power calculation. However, it is possible that, in reality, our found effect size may be smaller than the one found in the abovementioned study.

In summary, this study aims to examine the efficacy of the Karman Line — Tempo module for individuals with ABI who experience slowed IPS, focusing on the generalisation of compensatory strategies in everyday life situations. This study could contribute to upgrade the treatment of slowed IPS in cognitive rehabilitation and, more generally, improve our insight into implementing digital platforms in cognitive training.

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Contributors ACA developed the intervention and designed the study, and wrote and revised the manuscript draft in collaboration with the other co-authors. DB, RPCK, IAB and LF contributed to the study design and reviewed and extended the manuscript. IAB provided support in the set-up of the analysis of the study. All authors critically reviewed the content and approved the final manuscript. All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Competing interests None declared.

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